APPLICATIONS

Asset twins

Module 9 · Pavements, bridges and tunnels

An asset twin



Measuring condition

- Pavements: roughness (IRI), cracking, rutting
- Bridges: strain, vibration, deflection, inspections
- Drones and AI to find cracks

Predicting deterioration

- Statistical and Markov models for networks
- Mechanistic models for pavements
- Finite-element models updated with sensor data

Decisions

Without a twin

Fixed inspection cycles

Repair when visibly failed

With a twin

Risk-based inspection

Repair at the cheapest point in the lifecycle

Key standard

ISO 55000

Asset management: decisions linked to value, risk and cost